LESSON PLAN

ENGINE HEALTH MONITORING (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Cable identification, specifications, replacement, swaged type terminals splicing, swaged terminals splicing, swaged ball terminals, NICO press process and thimble eye
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Cable identification, specifications, replacement, swaged type terminals splicing, swaged terminals splicing, swaged ball terminals, NICO press process and thimble eye.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe the use of various standards, identification of bolts, nuts, safety methods and maintenance of cables.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of NICO press process, swaged terminal splicing, and thimble eye installation with safety precautions and instructor supervision.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary safety methods for handling aircraft cables?
(b) How do you identify standard cable specifications?
(c) What is the basic difference between swaged and NICO terminals?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Cable identification and specifications • Safety methods and bolt/nut standards • Cable replacement procedures	L & Ds	PPT, IB	15 min
(b)	• Swaged type terminals splicing • Swaged ball terminals installation • Thimble eye preparation	L & Ds	PPT, PDF	20 min
(c)	• NICO press process operation • Terminal crimping verification • Final inspection criteria	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the critical steps in the NICO press process.
(b) Describe the safety precautions required during cable replacement.
(c) How do you verify the integrity of a swaged terminal?

12. Home Assignment and Follow Up	Prepare a brief report on the differences between Bowden cable and Teleflex construction to be discussed in LP 05.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Cable system inspection, cable maintenance, fittings, pulleys and wear patterns, Construction and servicing of Bowden cable and Teleflex controls	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min